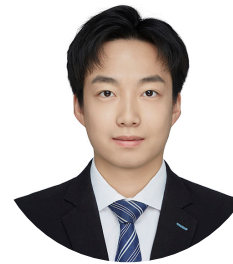


Wenhao Li

 Sereinme |  +86 15681220023 |  liwh23@mails.tsinghua.edu.cn |  liwh01 |
 0009-0004-7752-6737 |  Wenhao-Li-66 |  PhD in Electronic Engineering



Ph.D. candidate in Information and Communication Engineering at Tsinghua University (Class of 2023), working on GNSS positioning and time-synchronization security with 3 first-author papers including IEEE Transactions on Smart Grid. Hands-on across embedded FPGA development, data science, and embodied intelligence, with a record of taking algorithms from theory to validated systems.

Education



Ph.D. in Information and Communication Engineering

TSINGHUA UNIVERSITY

2023 - Present

Beijing, China

- Research focus: GNSS positioning and time-synchronization security — 3 first-author papers published or accepted, including IEEE Transactions on Smart Grid
- Courses: Functional Analysis | Matrix Analysis and Applications | Pattern Recognition | Introduction to Artificial Intelligence



Bachelor of Engineering in Electronic Science and Technology

TSINGHUA UNIVERSITY

2019 - 2023

Beijing, China

- Thesis: Research and Implementation of GNSS Data Compression Technology for Power Estimation Applications
- Courses: Database | Computer Networks | Principles of Communications | Data and Algorithms | Signals and Systems
- GPA: 3.78/4.0

Publications

- W. Li**, M. Zhong, M. Lu, and H. Li, "Full-channel asynchronous spoofing technique for GNSS time synchronization attack," *IEEE Trans. Smart Grid*, vol. 17, no. 2, pp. 1492–1502, Mar. 2026, doi: 10.1109/TSG.2025.3622566.
- W. Li**, M. Zhong, M. Lu, and H. Li, "Sequential separate clock drift anomaly detection method for GNSS time synchronization attack," in *Proceedings of the 39th International Technical Meeting of the Satellite Division of the Institute of Navigation (ION GNSS+ 2026)*, Orlando, Florida, Oct. 2026.
- W. Li**, L. Wang, M. Zhong, M. Lu, and H. Li, "An FD-DEFLATE data compression scheme for C/N0 estimation in GNSS interference monitoring," in *Proceedings of the 37th International Technical Meeting of the Satellite Division of the Institute of Navigation (ION GNSS+ 2024)*, Baltimore, Maryland, Oct. 2024, pp. 818–829. doi: 10.33012/2024.19871.
- M. Zhong, **W. Li**, M. Lu, and H. Li, "Feedback node aided distributed spoofing system for global navigation satellite system time synchronisation attack," *IET Radar Sonar Navig.*, vol. 19, no. 1, p. e70088, 2025, doi: 10.1049/rsn2.70088.
- Y. Ma, H. Li, Z. Zhou, Z. Wu, **W. Li**, and M. Lu, "A fast C/N0 estimation method based on the ratio of acquisition correlation value," in *China Satellite Navigation Conference (CSNC 2024) Proceedings*, C. Yang and J. Xie, Eds., Singapore: Springer Nature, 2024, pp. 567–577. doi: 10.1007/978-981-99-6932-6_47.

Professional Experience



Embodied Intelligence Algorithm Intern

SERES GROUP

2025.06 - 2025.08

Chongqing, China

- Targeted autonomous navigation for quadruped robots in unknown environments: surveyed VLM/VLN routes, then implemented a multimodal-perception navigation algorithm with a scene knowledge graph (SKG) that extended reliable path planning into cluttered, previously unseen scenes
- Diagnosed the root cause of parking-accuracy loss in the APA perception module by mapping how positioning error propagates through the control loop, then applied ESKF multi-sensor fusion with graph optimization to tighten pose estimation, improving parking accuracy and system robustness

Vision Language Navigation Automated Parking Assist



Algorithm R&D Intern

NAVITECH

2024.06 - 2024.08

Beijing, China

- Completed embedded development on a Xilinx FPGA platform in C, implementing and optimizing the hardware-software communication link to improve system response efficiency
- Designed a multi-peak signal capture module for unstable signal acquisition in complex scenarios, then ran robustness testing and performance evaluation that confirmed algorithm stability

Embedded Development Baseband Signal Processing

- Analyzed and mined large-scale datasets with SQL and Python, designing and validating regression prediction metrics that supported business decisions
 - Evaluated software functionality and practicality and optimized core performance indicators, improving overall system operating efficiency
- StatisticsData Processing

Projects

Vision-Language Navigation Application Research

SERES GROUP

2025.06 - 2025.08

Chongqing, China

- Addressed robot autonomy in unknown environments by researching VLM-based embodied navigation and building a scene knowledge graph (SKG) to strengthen path planning in complex scenes
 - Designed and implemented a multimodal feature-fusion strategy coupling visual perception with language understanding, enabling autonomous navigation and task execution for robots/agents without prior maps
- VLNVLMCVEmbodied AISKG

APA Positioning Error Propagation Research for Automated Parking System

SERES GROUP

2025.06 - 2025.08

Chongqing, China

- Investigated APA positioning accuracy by decomposing the error sources in the perception module and modeling how they propagate through the control system
 - Applied ESKF multi-sensor fusion with graph optimization to improve pose-estimation accuracy, giving parking path planning a more reliable positioning basis
- CVESKFGraph OptimizationAutonomous Driving

FPGA-based Streaming Data Compression Platform

PERSONAL PROJECT

2023.06 - 2024.04

Beijing, China

- Built an efficient streaming data compression architecture on an FPGA platform in Verilog
 - Designed a pipelined data-processing module that achieved low-latency, high-throughput compression of real-time data streams
- CompressionVerilogFPGADEFLATEZstandard

Subspace-based Face Recognition Technology

COURSE PROJECT

2024.05 - 2024.06

Beijing, China

- Studied subspace-based face representation and recognition, implementing dimensionality-reduction algorithms including SRC and Laplacian Eigenmaps
 - Constructed subspace models of face images to explore robust feature extraction and classification under varying illumination, expression, and occlusion
- SubspaceSRCFace IdentificationLaplacian

Audio-Video Joint Text Transcription Problem

COURSE PROJECT

2023.12 - 2024.01

Beijing, China

- Tackled joint recognition and transcription of audio-video bimodal signals by implementing multimodal information fusion
 - Extracted visual features with ResNet and modeled text sequences with BPE, improving speech recognition accuracy in complex scenarios
- BPEDLPythonResNet

Cocktail Party Effect Speaker Recognition Technology

COURSE PROJECT

2021.01 - 2021.02

Beijing, China

- Studied speaker recognition in complex acoustic environments by simulating target-speech separation and identification under the cocktail party effect
 - Used facial visual information to assist speech recognition, achieving accurate identity recognition in multi-speaker scenarios
- CVDLPythonSpeaker RecognitionFace Identification

Campus Experience

Deputy Secretary of the General Branch

DEPARTMENT OF ELECTRONIC ENGINEERING, TSINGHUA UNIVERSITY

2023.09 - 2024.06

Beijing, China

- Planned and coordinated external exchange activities for the Graduate Student League Committee and organized multiple department-level events, strengthening the committee's influence and cohesion

- Managed the full cycle of certification, graduation, and presentation activities for the Electronic Engineering Doctoral Lecturers Group, keeping operations efficient and orderly
- Connected with schools, universities, and enterprises inside and outside Tsinghua to expand presentation collaboration channels and raise the group’s social impact

- Organized and led the department’s practice team to Wuhan for in-depth field research and practical activities
- Won the department’s only gold award and placed in the top 4% university-wide (6/150), earning the “Excellent Team Leader” title

- Graded assignments and lab reports and ran regular Q&A sessions, helping the instructor refine teaching content and methods
- Compiled solution approaches and implementations for the Unscented Kalman Filter and Particle Filter, deepening students’ grasp of nonlinear non-Gaussian signal processing

- Graded assignments and tutored after class, explaining core algorithm principles to strengthen students’ application of statistical inference methods
- Co-authored exercises and case studies on the EM Algorithm, Kalman Filter, and Generalized Likelihood Ratio Test (GLRT), reinforcing the link between theory and practice

Certificates & Awards

- 2025Tsinghua University Outstanding Teaching Assistant Award
- 2024Tsinghua University Graduate Comprehensive Excellence Award
- 2024Tsinghua University Outstanding Communist Youth League Member
- 2022Tsinghua University Undergraduate Academic Excellence Award

Skills

Programming	Python MATLAB C C# Verilog SQL Java Rust
Positioning	GNSS RTK SLAM ESKF Graph Optimization
AI & Perception	VLN VLM CV Multimodal Perception SKG
Operating Systems	Linux Windows macOS
Languages	Chinese English Japanese